Presentation Script and Evidence Notes

Re-evaluating Multimodal LLMs for Influenza Forecasting

A 28-Model Benchmark of Accuracy, Peak Skill, Parameter Efficiency and Statistical Evidence

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The English script has 606 whitespace-delimited words and a planned duration of 5:00. Read only the script paragraphs during the timed talk. Follow-up answers, paper explanations and references are preparation material outside that budget. Human rehearsal has not been executed.

Scientific results retain the audited historical snapshot. Execution counts are dated to the owner handoff at 5 November 2026 00:15 HKT, not a live cluster reading. This edition changes the preparation language and expands the cited explanations; it adds no experiment result.

Slide	Time	Topic
1	0:00–0:15	Re-evaluating Multimodal LLMs for Influenza Forecasting
2	0:15–0:40	Four evaluation axes ask different questions
3	0:40–1:05	Reproduction and benchmarking use separate run sets
4	1:05–1:35	Reproduction differs by horizon and error measure
5	1:35–2:00	MIFlu leads the observed average-error ranking
6	2:00–2:25	Peak skill separates timing from height
7	2:25–2:50	Equal pass counts hide different passing cells
8	2:50–3:15	The parameter frontier exposes a size–error trade-off
9	3:15–3:45	Correct Holm supports no familywise rejection
10	3:45–4:05	Frozen-feature encoders underperform in this protocol
11	4:05–4:25	CPU tests confirm the inspected no-text invariant
12	4:25–4:45	The ten-seed expansion remains in progress
13	4:45–5:00	Model choice depends on the evaluation objective

Slide 1 Re-evaluating Multimodal LLMs for Influenza Forecasting

Planned timing 0:00–0:15 (15 seconds)

Spoken script

My project re-evaluates multimodal LLMs for influenza forecasting. I examine MIFlu and compare twenty-eight implementations. The question is how model choice changes when average accuracy, peak errors, model size and statistical evidence are considered together.

Likely follow-up

What is the contribution beyond reproduction?

A four-axis re-evaluation with an explicit statistical correction and implementation/provenance limits. Broad negative-LLM findings already exist; this is the specific influenza audit, not a first-ever claim.

Paper basis and interpretation

Simple-linear and LLM-removal critiques predate this project [4, 13]; Qiu et al.'s preprint reports conditional gains in larger cross-dataset settings [14]. Our narrower contribution connects audited influenza accuracy, peak endpoints, a cost subset and corrected inference. O1 checks persistence across origins/regions; Tashman motivates rolling-origin evaluation [23]. O2 adds validation-fitted height adjustment only if suitable prediction exports exist. A new mechanism cannot replace those checks or establish a universal verdict about LLM forecasting.

Reading locator: Zeng et al. [4], §4; Tan et al. [13], §3.2; Qiu et al. [14], abstract and §§3–5 (preprint); Tashman [23], publisher-indexed abstract.

Local evidence

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

02_数字表格/BENCH_PTE_PIE_BY_STEP.csv

audit/CLAIM_AUDIT.md

Slide 2 Four evaluation axes ask different questions

Planned timing 0:15–0:40 (25 seconds)

Spoken script

I use four evaluation axes. Accuracy measures overall prediction error. Peak skill measures timing and height. Parameter efficiency compares model size with error. Statistical evidence asks what differences the test design can support. Their coverage differs: twenty-eight, twenty-eight, eleven and seventeen models. Those denominators must stay separate.

Likely follow-up

Why do the denominators differ?

The source exports have different membership. Accuracy/peak cover 28 retained arms. Parameter data cover 11, excluding MIFlu and encoders. Paired comparisons cover 17 including MIFlu, excluding DLinear and encoders.

Paper basis and interpretation

The families test different explanations: simple temporal mapping [4], patches and inverted variable tokens [3, 5], and explicit cross-modal alignment [7, 8]. This motivates diverse controls, not a claim that 28 is an optimal count. It also makes implementation fidelity important: a named adaptation may omit a paper's decisive alignment or training component. DLinear is a relevant simple control in this literature. Its appearance in the source paper is not itself a scientific reason to omit it from a sensitivity analysis.

Reading locator: Zeng et al. [4], §4; PatchTST [3], §3.1; iTransformer [5], §3; CALF [7] and TimeCMA [8], author abstracts.

Local evidence

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

02_数字表格/BENCH_PTE_PIE_BY_STEP.csv

02_数字表格/BENCH_EFFICIENCY.csv

02_数字表格/BENCH_PAIRED_TESTS.csv

Slide 3 Reproduction and benchmarking use separate run sets

Planned timing 0:40–1:05 (25 seconds)

Spoken script

The historical reproduction uses ten seeds, while the wider benchmark uses three under shared cropped windows. Its twenty-eight implementations comprise thirteen conventional models, four LLM-based models and eleven frozen encoders. A new ten-seed expansion is underway. I keep its execution status separate from these historical results.

Likely follow-up

Why three seeds rather than ten for every model?

The retained benchmark is 28 × 4 × 3 = 336 cells. Three reflects the compute budget. More seeds can improve uncertainty; no adequate-power or no-benefit claim follows.

Paper basis and interpretation

The original three-seed citation is incorrect. PatchTST uses fixed-seed main results plus a five-seed supervised robustness check [3, App. A.6.1]; iTransformer reports five-run standard deviations [5, App. A.2, Table 5]. Neither passage establishes three seeds as their shared convention. Our choice is a compute compromise. Bouthillier et al. distinguish repeated-run precision from broader benchmark variation [19]. Extra seeds can improve the former; extra seasons/origins address different uncertainty. Ten reproduction repetitions do not make the separate three-seed benchmark equally precise.

Reading locator: PatchTST [3], Appendix A.6.1; iTransformer [5], Appendix A.2 and Table 5; Bouthillier et al. [19], Appendix C.3.

Local evidence

02_数字表格/ESSAY_VS_REPRODUCTION_TABLE_V.csv

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

05_程序代码/eval_windows.py

handoffs/20261105_return/RETURN_SUMMARY.md

audit/return_review_20261105/RAW_RESCORE_SUMMARY.json

audit/return_review_20261105/SUPPLIED_no_text_runtime.xml

Slide 4 Reproduction differs by horizon and error measure

Planned timing 1:05–1:35 (30 seconds)

Spoken script

The reproduction’s mean MSE is four point five percent higher than the paper table. It is lower at horizons twenty-four and thirty-six, but higher at forty-eight and sixty. MAE is seven point nine percent higher overall, with only one lower horizon. The two error measures need separate explanations, and the causes of the gaps remain unresolved.

Likely follow-up

Does the figure prove an exact reproduction?

No. The plotted means are verified against the supplied reproduction CSV. Raw ten-seed predictions/logs and a pinned author revision remain unavailable for independent complete reproduction.

Paper basis and interpretation

Repeated-run estimates and traceable evaluation address different weaknesses [19, 22]. The oracle checks the arithmetic of a transcribed record; saved predictions would check whether the pipeline produced it. The practical chain is run configuration → prediction → scoring code → cell → aggregate. A benchmark plot is another summary and cannot establish a ten-seed reproduction trajectory. Missing links limit the claim even when the mean is close. The imported panels remain useful illustrations if their true provenance is stated.

Reading locator: Bouthillier et al. [19], variance analysis; Kapoor and Narayanan [22], model info sheets.

Local evidence

02_数字表格/ESSAY_VS_REPRODUCTION_TABLE_V.csv

audit/NUMERICAL_AUDIT.md

Slide 5 MIFlu leads the observed average-error ranking

Planned timing 1:35–2:00 (25 seconds)

Spoken script

MIFlu has the lowest observed mean MSE, about one point five five seven four. PatchTST and NLinear follow in that order, and MAE gives the same leading three. This comparison concerns project implementations on one national split. It does not show that the multimodal component caused MIFlu's advantage.

Likely follow-up

Are these the original algorithms with verified identical implementation?

They are project implementations/adaptations. Their exact equivalence to each published comparator has not been independently established. The ranking alone cannot prove method fidelity.

Paper basis and interpretation

The families test different explanations: simple temporal mapping [4], patches and inverted variable tokens [3, 5], and explicit cross-modal alignment [7, 8]. This motivates diverse controls, not a claim that 28 is an optimal count. It also makes implementation fidelity important: a named adaptation may omit a paper's decisive alignment or training component. DLinear is a relevant simple control in this literature. Its appearance in the source paper is not itself a scientific reason to omit it from a sensitivity analysis.

Reading locator: Zeng et al. [4], §4; PatchTST [3], §3.1; iTransformer [5], §3; CALF [7] and TimeCMA [8], author abstracts.

Local evidence

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

Slide 6 Peak skill separates timing from height

Planned timing 2:00–2:25 (25 seconds)

Spoken script

Peak errors use ILITOTAL in patient counts, rather than the seven-channel standardised loss. The joint screen requires timing error within one week and height error within twenty-five percent in the same aggregated cell. MIFlu and NLinear each pass four of sixteen cells. These are horizon–lead combinations, not independent seasons.

Likely follow-up

Is 4/16 an estimated 25% season-success probability?

No. There are four horizons times four fixed leads. The errors are aggregated within each cell before applying thresholds. Event-level success probabilities require a different analysis.

Paper basis and interpretation

Kramer and Shaman evaluate forecast-level peak accuracy by lead time [2]. Our sixteen configuration cells have a different denominator. Applying a threshold to an average is mathematically different from averaging event-level pass indicators. Thus 4/16 cannot be relabelled as season success probability. Bracher et al. discuss calibration of predictive distributions [18, §4.2]; small peak-height error does not establish it. Estimating event success requires retaining individual outcomes, defining the event denominator and accounting for dependence.

Reading locator: Kramer and Shaman [2], Forecast accuracy and comparison / Fig. 2; Bracher et al. [18], §4.2.

Local evidence

02_数字表格/BENCH_PTE_PIE_BY_STEP.csv

audit/LITERATURE_REVIEW.md source [2]

Slide 7 Equal pass counts hide different passing cells

Planned timing 2:25–2:50 (25 seconds)

Spoken script

The grid shows different passing locations. NLinear passes the first lead at all four horizons. MIFlu passes three first-lead cells and the second lead at horizon forty-eight. Across all cells, NLinear has lower peak-height error, while MIFlu has better timing. The leading-three ordering is inverted at both ends between MSE and peak-height error.

Likely follow-up

Does this demonstrate peak-skill decoupling?

It demonstrates local objective-dependent rankings and a timing–height trade-off. Overall correlations remain +0.736 for MSE–PIE and −0.429 for MSE–pass count. Independence or universal decoupling is not established.

Paper basis and interpretation

Zeng et al. describe NLinear's last-value subtraction, linear map and addition back [4]. This gives a testable hypothesis about level handling, not a proven explanation for our peaks. The ranking mismatch itself follows a simpler distinction: MSE aggregates many points and seven channels, whereas PIE concentrates on detected ILITOTAL peak height [21 gives the error-measure context]. A model can improve many non-peak points yet worsen this endpoint. Controlled variants and peak/residual diagnostics are needed to attribute the observed trade-off to architecture.

Reading locator: Zeng et al. [4], §4 NLinear; Hyndman and Koehler [21], §2.1.

Local evidence

02_数字表格/BENCH_PTE_PIE_BY_STEP.csv

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

Slide 8 The parameter frontier exposes a size–error trade-off

Planned timing 2:50–3:15 (25 seconds)

Spoken script

The parameter frontier covers eleven exported arms. NLinear is smaller, while PatchTST has lower MSE. PatchTST uses about two hundred and eleven times as many parameters for eleven point two percent lower observed MSE. The size fit explains only about two percent of observed MSE variation. Parameter count remains a size proxy, with runtime unmeasured.

Likely follow-up

Why is MIFlu absent, despite the study title mentioning LLMs?

Its parameter count is not in the supplied efficiency export. Missing values remain missing. The full benchmark has 28 arms, but this measured subset has 11.

Paper basis and interpretation

iTransformer evaluates error alongside speed and memory [5, App. D], a precedent for considering performance with cost. Our cost axis is narrower: exported parameters. The two local frontier members follow from pairwise dominance in the eleven-model subset, not a paper's recommendation of a universal winner. A Pareto graph removes dominated choices; the remaining choices still need a budget or acceptable-error preference. Missing MIFlu cost data prevents calling it the frontier of all 28 models.

Reading locator: iTransformer [5], Appendix D Model Efficiency; local BENCH_EFFICIENCY.csv.

Local evidence

02_数字表格/BENCH_EFFICIENCY.csv

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

Slide 9 Correct Holm supports no familywise rejection

Planned timing 3:15–3:45 (30 seconds)

Spoken script

The paired family has seventeen models, including MIFlu. Proper Holm rejects zero of one hundred and thirty-six pairs. The NLinear–PatchTST p-value is zero point one one seven two. Coarse exact-test resolution already blocks the first Holm threshold, and dependent cells add concern. The observed gain remains. Equivalence is unproven.

Likely follow-up

Why is the minimum adjusted p already above 0.05?

At n=12 the documented two-sided exact raw-p floor is 2/4096. Multiplying by the first Holm factor 136 gives about 0.0664. This family cannot pass the first threshold at that resolution. Dependence is a separate validity concern.

Paper basis and interpretation

Demšar describes starting with the smallest p and stopping at the first failed Holm threshold [20]. Our first threshold is 0.05/136. The documented two-sided twelve-contribution sign-flip floor is 2/2^12; multiplying by 136 gives 0.06640625, so no pair can start the rejection sequence. This floor is local mathematical reasoning, not a result reported by Demšar for influenza. Holm permits dependence among valid p-values, but cannot repair invalid raw p-values from unjustified within-pair exchangeability. These are separate limitations.

Reading locator: Demšar [20], §3.2.2 Holm step-down; local exact-enumeration derivation.

Local evidence

02_数字表格/BENCH_PAIRED_TESTS.csv

audit/HOLM_RECOMPUTATION.csv

Slide 10 Frozen-feature encoders underperform in this protocol

Planned timing 3:45–4:05 (20 seconds)

Spoken script

Eleven frozen encoders with a linear readout all perform worse than the decision tree and pass zero joint cells. This is a negative result for the tested feature pipeline. Generation, prompting, fine-tuning and numerical time-series foundation models remain outside this test.

Likely follow-up

Does this prove that all LLM forecasting is ineffective?

No. It is restricted to the frozen-feature/readout protocol. Broader LLM-based and numerical foundation methods require their own faithful comparisons and controls.

Paper basis and interpretation

Time-LLM freezes its LLM but adds learned reprogramming and a prompt prefix [6]. Thus “frozen” alone does not define the same experiment as our feature/linear-readout pipeline. Chronos instead learns forecasting from numerical sequences [9]. These method differences justify keeping our negative finding narrow. To isolate a readout bottleneck, vary the readout/alignment while holding the encoder fixed; to isolate pretraining, compare matched pretrained and random backbones. Our eleven-arm result does not make those controlled separations.

Reading locator: Time-LLM [6], author abstract; Chronos [9], §3.

Local evidence

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

02_数字表格/BENCH_PTE_PIE_BY_STEP.csv

Slide 11 CPU tests confirm the inspected no-text invariant

Planned timing 4:05–4:25 (20 seconds)

Spoken script

The returned CPU log records three no-text assertions passed, with zero skips, under fixed-weight evaluation. This supports the inspected causal-mask explanation. It does not establish training-time text effects. The ablation queue reports sixty of sixty cells done, with per-variant analysis pending.

Likely follow-up

What do the three passing CPU tests establish?

They support the three named invariants for fixed weights, matched inputs and CPU evaluation in the supplied execution record. They do not establish training-time text/LoRA effects, all fusion paths or equality of independent training trajectories. Whole-archive collection still fails.

Paper basis and interpretation

The Transformer paper explains masking that prevents attention to subsequent positions [24, §3.1]. With the inspected ordering, time positions cannot acquire later text through these layers, and the head reads time positions. This is a structural argument about a fixed forward path, not certification of repository runtime or bitwise equality. The three archived assertions must control weights, inputs and evaluation randomness when comparing changed text and matched full/no-text outputs. The supplied JUnit now records three CPU assertions passed with zero skips; they were not rerun in this review. Independent training can consume randomness differently and requires its own test.

Reading locator: Vaswani et al. [24], §3.1 Decoder; local ForecastingLLM.forward and archived assertions.

Local evidence

05_程序代码/miflu_model.py

tests_archive/test_miflu_ablations.py

audit/NO_TEXT_VERIFICATION.md

handoffs/20261105_return/RETURN_SUMMARY.md

audit/return_review_20261105/RAW_RESCORE_SUMMARY.json

audit/return_review_20261105/SUPPLIED_no_text_runtime.xml

Slide 12 The ten-seed expansion remains in progress

Planned timing 4:25–4:45 (20 seconds)

Spoken script

Core outputs are complete for one hundred and sixty cells. The national expansion remains unfinished: the package records one hundred and seventeen, followed by an owner report of one hundred and forty-one. Scoring windows still differ. O1 prioritises harmonisation, regional validation and stronger controls.

Likely follow-up

Does 160/160 mean the new benchmark is ready to publish?

It establishes core artefact completeness. Stored MIFlu scores use cropped windows while the other supplied models use all valid windows, and complete seven-channel MIFlu exports are missing. The national 1120 matrix, harmonised scoring and external-validity evidence remain incomplete.

Paper basis and interpretation

Kapoor and Narayanan ask for explicit reporting arguments rather than equating code availability with a valid result [22]. Here distinguish document rebuilding, CSV arithmetic, raw prediction rescoring and model retraining. A supervisor can begin with the CSV audit; the full bundle supports one raw-cell smoke check. Model text invariance needs the three named runtime assertions without skips in a pinned environment. The useful deliverable is a traceable evidence chain with exact commands and status. Passing one evidence level does not certify all the others.

Reading locator: Kapoor and Narayanan [22], model info sheets; EXECUTION_STATUS.md and NO_TEXT_VERIFICATION.md.

Local evidence

audit/INNOVATION_DECISION.md

audit/REPO_AUDIT.md

audit/GRADE_READINESS.md

handoffs/20261105_return/RETURN_SUMMARY.md

audit/return_review_20261105/RAW_RESCORE_SUMMARY.json

audit/return_review_20261105/SUPPLIED_no_text_runtime.xml

Slide 13 Model choice depends on the evaluation objective

Planned timing 4:45–5:00 (15 seconds)

Spoken script

The historical results show different choices for average error, peak height and parameter count. The new run must be scored consistently before changing those conclusions. I propose completing that comparison and validating across origins and regions. O1 remains the next supervision decision.

Likely follow-up

What can you confidently defend today?

The exported arithmetic, bounded local trade-offs and proper Holm correction. The internal report Q01–Q25 supplies the explanations and evidence locators. Deployment value and complete independent reproduction remain unestablished.

Paper basis and interpretation

Kapoor and Narayanan ask for explicit reporting arguments rather than equating code availability with a valid result [22]. Here distinguish document rebuilding, CSV arithmetic, raw prediction rescoring and model retraining. A supervisor can begin with the CSV audit; the full bundle supports one raw-cell smoke check. Model text invariance needs the three named runtime assertions without skips in a pinned environment. The useful deliverable is a traceable evidence chain with exact commands and status. Passing one evidence level does not certify all the others.

Reading locator: Kapoor and Narayanan [22], model info sheets; EXECUTION_STATUS.md and NO_TEXT_VERIFICATION.md.

Local evidence

02_数字表格/BENCH_MSE_BY_MODEL_L.csv

02_数字表格/BENCH_PTE_PIE_BY_STEP.csv

02_数字表格/BENCH_PAIRED_TESTS.csv

Requirements_ProjectProgressReport_2627A (1).pdf

handoffs/20261105_return/RETURN_SUMMARY.md

audit/return_review_20261105/RAW_RESCORE_SUMMARY.json

audit/return_review_20261105/SUPPLIED_no_text_runtime.xml

References for preparation

Reference numbers follow the internal report and slide preparation notes. The rewritten journal manuscript has its own citation order.

[2] Sarah C. Kramer and Jeffrey Shaman, “Development and validation of influenza forecasting for 64 temperate and tropical countries,” PLOS Computational Biology 15(2), e1006742, 2019. https://doi.org/10.1371/journal.pcbi.1006742

[3] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong and Jayant Kalagnanam, “A Time Series is Worth 64 Words: Long-term Forecasting with Transformers,” ICLR, 2023. https://arxiv.org/abs/2211.14730

[4] Ailing Zeng, Muxi Chen, Lei Zhang and Qiang Xu, “Are Transformers Effective for Time Series Forecasting?,” AAAI (preprint record opened), 2023. https://arxiv.org/abs/2205.13504

[5] Yong Liu et al., “iTransformer: Inverted Transformers Are Effective for Time Series Forecasting,” ICLR (preprint record opened), 2024. https://arxiv.org/abs/2310.06625

[6] Ming Jin et al., “Time-LLM: Time Series Forecasting by Reprogramming Large Language Models,” ICLR, 2024. https://arxiv.org/abs/2310.01728

[7] Peiyuan Liu et al., “CALF: Aligning LLMs for Time Series Forecasting via Cross-modal Fine-Tuning,” arXiv:2403.07300, 2024. https://arxiv.org/abs/2403.07300

[8] Chenxi Liu et al., “TimeCMA: Towards LLM-Empowered Multivariate Time Series Forecasting via Cross-Modality Alignment,” AAAI, 2025. https://arxiv.org/abs/2406.01638

[9] Abdul Fatir Ansari et al., “Chronos: Learning the Language of Time Series,” Transactions on Machine Learning Research, 2024. https://arxiv.org/abs/2403.07815

[13] Mingtian Tan, Mike A. Merrill, Vinayak Gupta, Tim Althoff and Thomas Hartvigsen, “Are Language Models Actually Useful for Time Series Forecasting?,” NeurIPS 37, 2024. https://arxiv.org/abs/2406.16964

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[18] Johannes Bracher, Evan L. Ray, Tilmann Gneiting and Nicholas G. Reich, “Evaluating epidemic forecasts in an interval format,” PLOS Computational Biology 17(2), e1008618, 2021. https://doi.org/10.1371/journal.pcbi.1008618

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[20] Janez Demšar, “Statistical Comparisons of Classifiers over Multiple Data Sets,” Journal of Machine Learning Research 7, 1–30, 2006. https://jmlr.org/papers/volume7/demsar06a/demsar06a.pdf

[21] Rob J. Hyndman and Anne B. Koehler, “Another look at measures of forecast accuracy,” International Journal of Forecasting 22(4), 679–688, 2006. https://doi.org/10.1016/j.ijforecast.2006.03.001

[22] Sayash Kapoor and Arvind Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” Patterns 4, article 100804, 2023. https://doi.org/10.1016/j.patter.2023.100804

[23] Leonard J. Tashman, “Out-of-sample tests of forecasting accuracy: an analysis and review,” International Journal of Forecasting 16(4), 437–450, 2000. https://doi.org/10.1016/S0169-2070(00)00065-0

[24] Ashish Vaswani et al., “Attention Is All You Need,” Advances in Neural Information Processing Systems 30, 2017. https://papers.neurips.cc/paper/7181-attention-is-all-you-need.pdf

Proposed V8 update (notes only; owner PPTX unchanged)

Replace the headline result with the new common national panel: MIFlu 1.7944 < PatchTST 1.9521 < NLinear 2.0782 (all-valid). Keep the revision-9.2 table explicitly labelled as the old frozen snapshot.

For peak skill, state the new step-level counts—MIFlu 252/1680, PatchTST 215/1680, NLinear 57/1680—and do not call NLinear the current champion.

Mention the legacy walk-forward result only with its separate five-fold, three-seed provenance. Describe Chronos/TimesFM as a separate zero-shot rolling-origin roster, not as 28-arm or ten-seed evidence.

State that ablation depth is non-monotonic, CDC vintage results are retrospective, resources are representative-subset measurements, and the exact JBHI template class remains unverified. Do not claim submission readiness or scientific certification.

Validator PASS is structural only (scientific_readiness: NOT_ASSESSED).